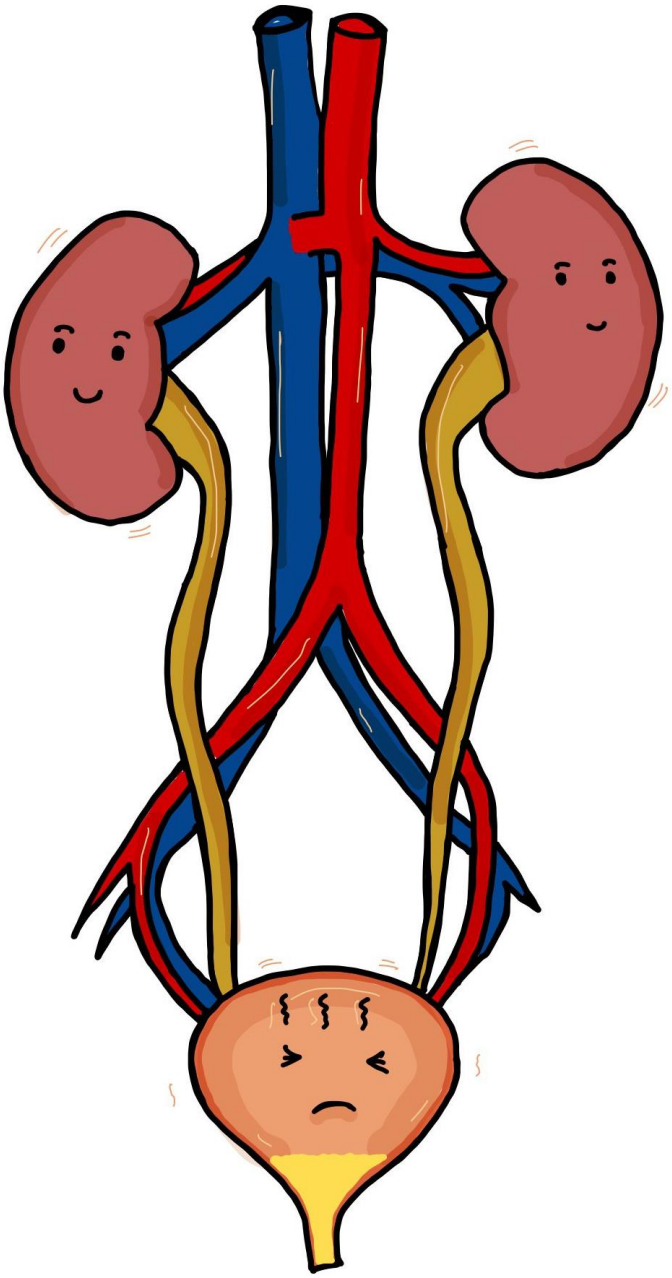
Pediatric nephrology



د. عبدالرحمن الجعكي

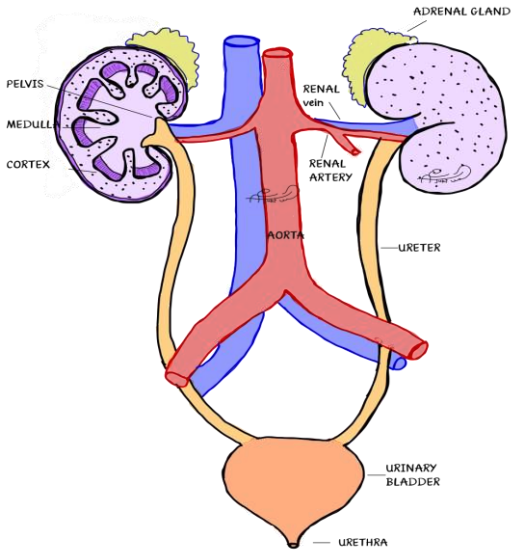
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"Introduction"

Renal Anatomy:

Kidney is formed of:

- Cortex (outer zone): Contains glomeruli, PCT, DCT & collecting ducts
- Medulla (inner zone): Contains loop of Henle, final portions of CDs ,It is formed of renal pyramids (renal papilla is tire apex of renal pyramid)
- Draining system for urine is formed of:
  - Collecting ducts, minor calyces, Major calyces, Renal pelvis
  - Ureters: urine passes along the ureter by peristalsis. It passes to the Urinary bladder & Urethra.



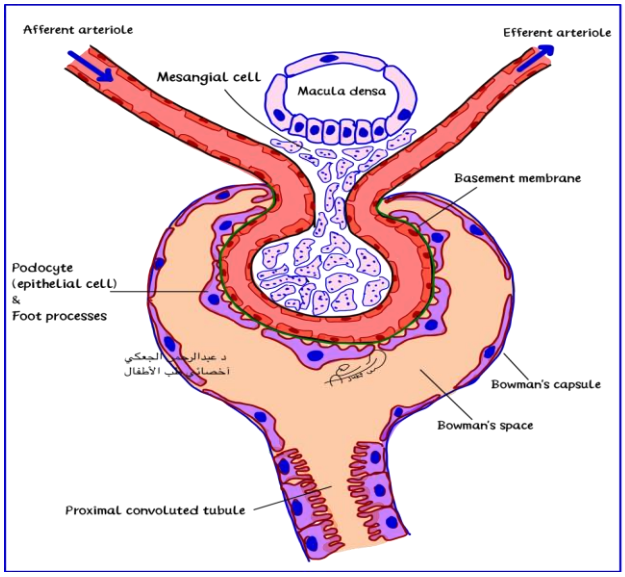
URINARY SYSTEM

The nephron:

Nephron is the **functional & structural unit of the kidney**. It is formed of:

1-Glomerulus: formed of

- Bowman's capsule: formed of visceral (Podocytes) & parietal layers .
- Tuft of capillary loops supplied by afferent arteriole & drained by
- a slightly smaller efferent arteriole
- Mesangium (: mesangial cells & mesangial matrix): lies between glomerular capillaries. It has supportive & immune functions



2-Tubule: PCT, loop of Henle (thin & thick segments) & DCT.

Glomerular Filtration barrier

formed of:

1. **Endothelial cells of glomerular capillaries:** fenestrated
2. **Capillary basement membrane:** formed of type IV collagen, proteoglycan.
3. **Epithelial cells of the visceral layer of Bowman's capsule (Podocytes):** podocytes ' have foot processes which are implanted on the glomerular BM leaving filtration slits

Filtration barrier is effective in preventing proteinuria due to:

1. **Size-selective properties:** fenestrations prevent passage of high MW substances
2. **Charge-selective properties:** -ve charges that repel proteins

Functions of kidney:

1. Regulation of fluid balance (Water homeostasis) & blood pressure
2. Regulation of electrolyte balance (Na , K, Ca, PO4)
3. Regulation of acid-base balance
4. Excretion of waste products
5. Vitamin D activation
6. Stimulation of erythropoiesis (Erythropoietin)

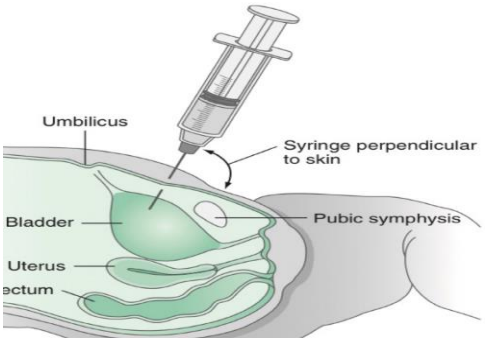
Glomerular Function:

- The kidney received up to 25% of cardiac output
- Glomerular filtration rate (GFR) is low in the newborn infant (especially low in preterm) ≃ 20-30 ml/min per 1.73m^2 and increasing gradually and reach adult value at age of 2 years (80-120 ml/min per 1.73m^2)
- Schwartz (GFR) = $\frac{\text{height(cm)} \times K}{\text{Serum cr (micromole/L)}}$ K = <1y =0.4, >1y=0.5, Adolescent =0.7

Investigation for kidney and urinary tract:

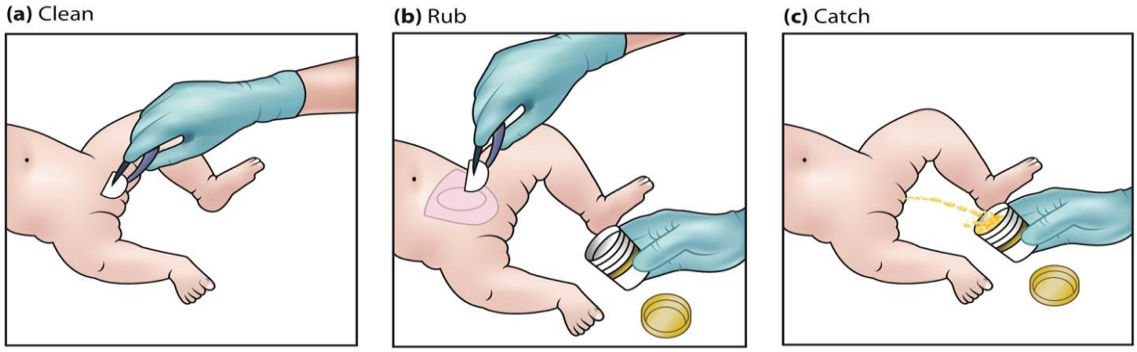
Collection of urine samples.

1–suprapubic aspiration (SPA): used in infants or newborn (bladder still abdominal organ) when a fine needle attached to a syringe is inserted directly into the bladder just above the symphysis pubis under ultrasound guidance/ it may be used in severely ill infants requiring urgent diagnosis and treatment, but it is an invasive procedure.

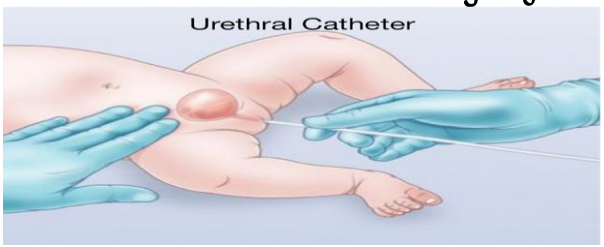




2–clean– catch : sample into a waiting clean container when the nappy is removed. This is the recommended method. The child can be encouraged to pass urine by gently rubbing their lower abdomen with gauze, soaked in cold water (sometimes called the 'Quick– wee technique.



3–midstream urine sample :it's a preferable method in school age children .
4– adhesive plastic bag: applied to the perineum after careful washing, although there may be contamination from the skin or faeces .
5–Urethral catheter: if there is urgency in obtaining a sample and no urine has been passed .





1. Urine for Routine examination: Urine Analysis

- Color: Normal urine Color is yellow (Due to urochrome)
    1. Dark Urine
        - Concentrated Urine (Dehydration)
        - Small amount of blood in urine
        - Obstructive jaundice

2. Cloudy (Turbid) Color : Urinary tract infection
3. Frothy Urine: Proteinuria
4. Colored Urine
   - Drugs – diet.
   - Black: Alkaptonuria
   - Red مهم
5. Light Colored Urine (Diluted) – DM, DI & RTA

Differential diagnosis of dark (Red) urine

1. **Hematuria (RBC in urine)**
2. Hemoglobinuria (acute hemolysis)
3. Methemoglobinemia (crushed ms injury)
4. Drugs as (rifampicin–sulfonamides –chloroquine–deferoxamine–metronidazole)
5. Diet containing dyes as (beets –blackberries)
6. Metabolites: as melanin –urates –homogentisic acid)

1. Smell: Offensive smell in "UTI" – some metabolic diseases give special urine odors as phenylketonuria –>musty odor)
2. Specific Gravity (Normal 1.015 – 1.025)
   - Low : Diabetes Insipidus , Excessive Hydration ,acute tubular necrosis pyelonephritis ,RF .
   - High : Dehydration ,SIADH , Pre renal ARF ,renal artery stenosis ,proteinuria
3. PH Normal (4.5 – 8)
   - High PH (Alkaline Urine) : UTI –Drugs (Amphotericin)– Systemic Alkalosis– Vegetarian Diet
   - Low PH (Acidic Urine): Starvation –Renal tubular acidosis –Systemic acidosis –PKU high protein diet
4. Nitrite – Produced by Bacterial reduction of nitrate
   - If +ve may Indicate UTI
   - If –ve it doesn't exclude UTI
5. Microscopic Examination
   1. RBC normal <5 Per HpF.

Definition of hematuria :–**Microscopic hematuria > 5 RBC HPF in 10 ml freshly voided centrifuged urine.(>10 RBC per HPF : illustrated textbook 6th edition)**

Types of Haematuria:

- Microscopic and Macroscopic = gross
- Painful or Painless Haematuria
- Initial, terminal or throughout the urine stream
- Glomerular or Extra glomerular Haematuria

	Glomerular Hematuria	Extra Glomerular Hematuria
Color	Cola or Tea Colored	Bright red in color
RBC Morphology	Dysmorphic	Isomorphic
RBC Casts	Present	Absent
Clots	–	+
Protein Urea	+ >100 mg/dl	–

Causes:–

Glomerular	Extra glomerular
• Glomerulonephritis as post strep GN • IgA nephropathy, Alpert syndrome • Henoch sholein purpura other vasculitis. • SLE Nephritis, HUS • SCA glomerulopathy • Other types of GN .	• UTI (**Most Common causes**) • Stones , Hypercalciuria. • Trauma • Tumor – Wilms • Drugs – bleeding disease, SCA. • Renal vein thrombosis • Rare infection as TB or schistosomiasis

2. WBC :– Normal <5 Per HpF
   DD of pyuria (WBC >5) usually indicate UTI "infection + culture & sensitivity positive

If there's (pyuria but C&S negative) it called **sterile pyuria** :–

1. Renal Stone
2. T B
3. Febrile illness
4. Dehydration
5. Appendicitis
6. Kawasaki disease –SLE

2. Cast Types
   - Hyaline casts – DD (Normal–Fever–Nephrotic syndrome–Heart failure)
   - **Red cell & Granular casts** Usually indicate **glomerular** disease
   - **WBC casts** – It usually indicates **pyelonephritis**.
3. Epithelial Cells: is normal finding

2–Urine Culture and sensitivity (diagnostic for UTI)

3–Imaging Study:

1. Ultra Sound Abdomen: Provides anatomical assessment but not function.
   , Excellent at visualizing urinary tract dilatation, stones, and nephrocalcinosis .

- **Advantage**:– Simple ,sensitive, Non–invasive,can be done in acute disease
- **Disadvantages**: operator dependent, may not detect all renal scars

2. Micturating Cystourethrography "MCU":(diagnosis of **VUR**)
   Contrast introduced into the bladder through urethral catheter can visualize bladder and urethral anatomy. Detects **vesicoureteric reflux (VUR)** and urethral obstruction
   Disadvantages: invasive and unpleasant investigation especially beyond infancy, high radiation dose, and can introduce infection .

Vesico ureteric reflux

Definition:

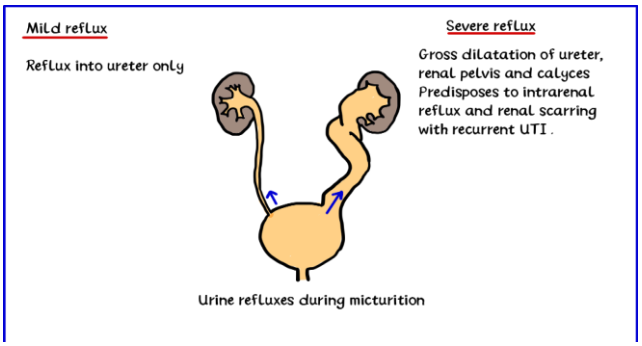
Retrograde flow of urine from bladder into upper urinary tract (ureter and renal pelvis) incidence 1%

Causes:

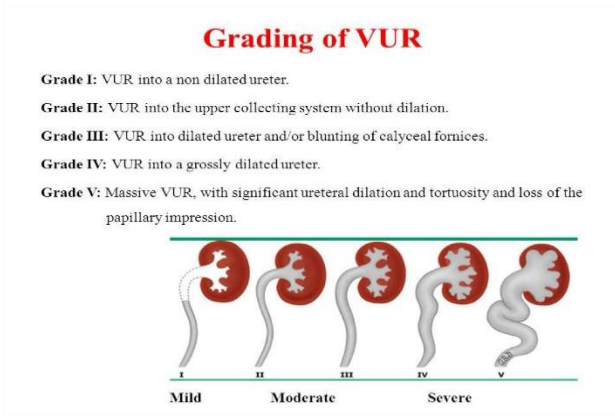
1. Congenital or primary VUR – Usually familial (30%–50% in first-degree relatives)
   - Is due to abnormality of vesico – ureteric junction
   - 50% of girls and 30% of boys have reflux at birth
   - It Is associated with :–Duplication of ureter –Ureteral ectopia –Urethrocel .
2. Secondary or acquired VUR as :–
   1. UTI as cystitis (Recurrent)
   2. Urinary obstruction – posterior urethral valve and neuropathic bladder
   3. Surgical procedure

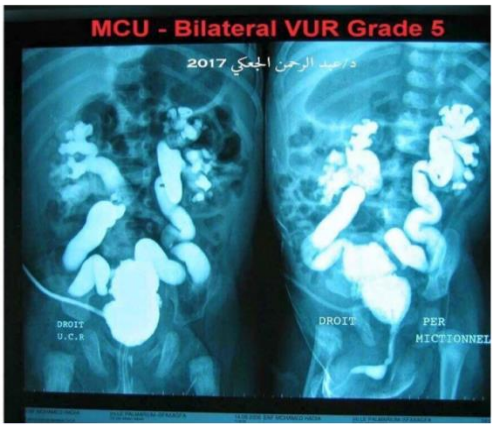
Pathophysiology & Complication of VUR:

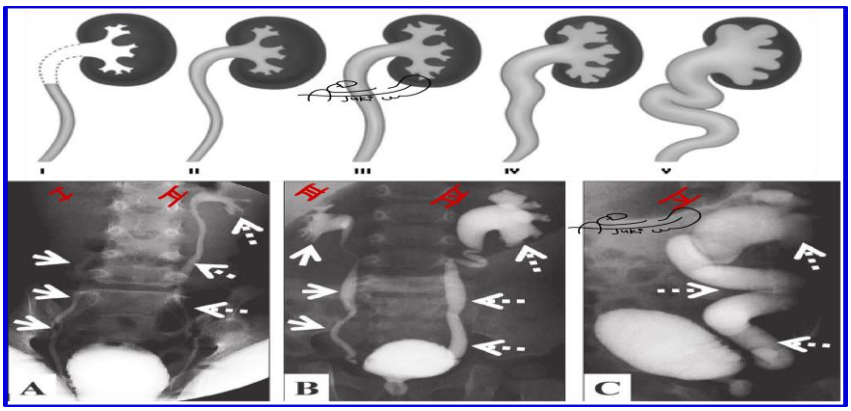
- Urine returning from ureters after micturation to blad → incomplete bladder emptying → (stasis) infection (cystitis) and may lead to → kidney infection pyelonephritis
- Bladder voiding pressure is transmitted to the renal pelvis & kidneys lead to renal tissue damage (Hydronephrosis)
- Recurrent pyelonephritis→renal damage→ Renal Scare →Shrunken, poorly functioning segment of kidney (reflux nephropathy)→if bilateral & severe →hypertension & CKD may develop.



Diagnosis : **Micturating cystourethrogram (MCUG)** :– for diagnosis & grading of VUR



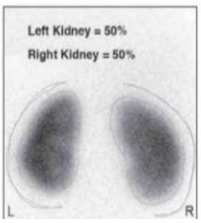


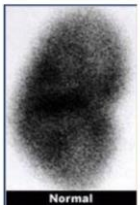


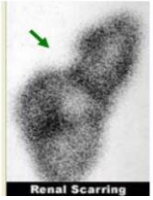
Treatment :–

- Grade I , II, III :may disappear spontaneously just using prophylactic antibiotics (Low–dose trimethoprim is used most often, but nitrofurantoin or cephalexin may be given.)
- Grade IV, V :Need surgical reimplantation of ureter or endoscopic injection of synthetic material at ureteric junction .

3. **DMSA (Di – mercapto – succinic – acid)**:
   - It's Static scan
   - Isotope injected IV and it taken up by and excreted by renal tubules
   - Indication : For Diagnosis of **renal scar** (areas of non–functioning renal tissue)







4. **X Ray KUB** : Identifies unsuspected spinal abnormalities
   - May identify renal stones, but poor at showing nephrocalcinosis
5. **DpTA – Technetium (MAG3 renogram)** :
   - It's dynamic scan
   - Absorbed by glomerular cells
   - Diagnosis of renal obstruction

6. **CT scan and MRI:**

4–Renal Biopsy: –is invasive procedure done in special condition to reach the final diagnosis (indication discussed later)

5–Blood Sample: –

- RFT: serum creatinine &Urea (Urea =2xBUN)
- CBC (anemia)
- Electrolytes (Na – K – CL – mg), serum ca, ph.
- ABG (PH – Hco3)
- Other investigations:as complement level, Total protein …

"Congenital Anomalies"

1– Renal agenesis:

Unilateral agenesis: – often is discovered accidentally or during evaluation for other anomalies.

Bilateral agenesis: – severe oligohydramnios resulting in Potter sequence, which is fatal due to failure of development of fetal lungs.

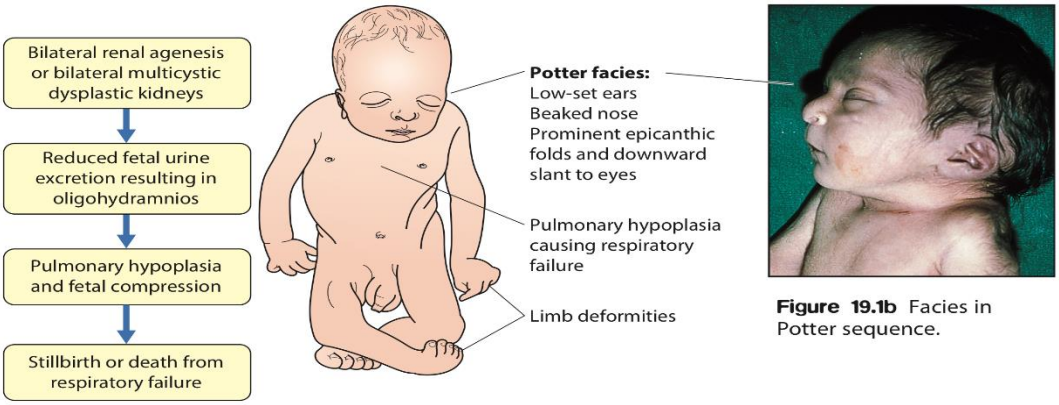
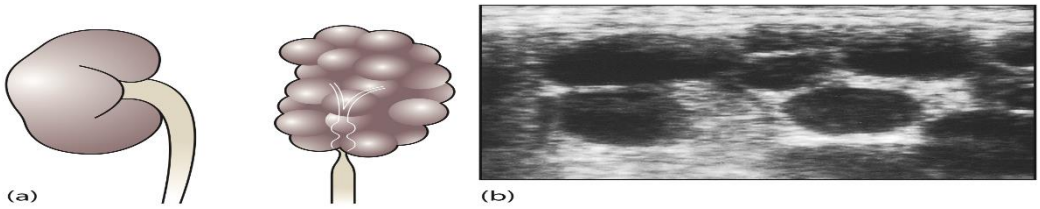


Figure 19.1b Facies in Potter sequence.

Multicystic dysplastic kidney (MCDK) – Results from the failure of union of the ureteric bud (which forms the ureter, pelvis, calyces, and collecting ducts) with the nephrogenic mesenchyme. It is a non–functioning structure with large fluid–filled cysts with no renal tissue and no connection with the bladder, Half will have involuted by 2 years of age, and nephrectomy is indicated only if it remains very large or hypertension develops, but this is rare. MCDKs do not produce urine and, if they are bilateral, Potter sequence will result.

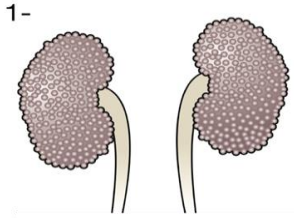


Polycystic Kidney

Two types : 1– Infantile type (AR)=**ARPKD** 2– Adult type (AD)=**ADPKD**

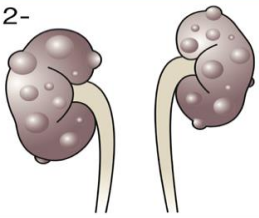
- **Infantile Polycystic kidney disease**
  - AR 1:10,000
  - Antenatal: H/O oligohydramnios & can be diagnosed by u/s
- **Post–natal C/P:**
  - Bilateral lion mass *.
  - HTN.
  - RD (2ry to pulmonary hypoplasia)
  - CRF (high Ur – high Cr)
  - Usually associated with congenital hepatic periportal fibrosis ,liver cirrhosis & portal HTN late childhood.
- **Rx:**
  - Control HTN.
  - Dialysis or Renal transplantation
  - Liver support.

1-



Autosomal recessive polycystic kidney disease (ARPKD). There is diffuse bilateral enlargement of both kidneys.

2-



Autosomal dominant polycystic kidney disease (ADPKD). There are bilateral separate cysts of varying size between normal renal parenchyma. The kidneys are enlarged.

Adult type: ADPKD has an incidence of 1 in 1000/ the main symptoms in childhood are hypertension, and it causes chronic kidney disease requiring renal replacement in late adulthood. It is associated with several extrarenal features including cysts in the liver and pancreas, cerebral aneurysms, Hge, and mitral valve prolapse.

Causes of palpable kidneys (Loin mass)

Unilateral

1. • Multicystic dysplastic kidney
2. • Compensatory hypertrophy of normal kidney
3. • Obstructed hydronephrosis
4. • Renal tumour (Wilms tumour), neuroblastoma
5. • Renal vein thrombosis

Bilateral

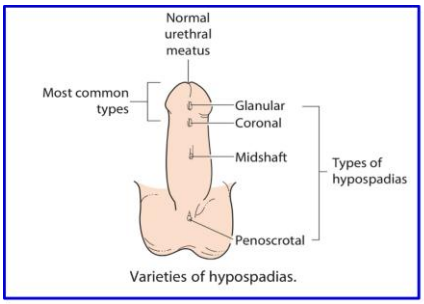
1. • Autosomal recessive polycystic kidneys
2. • Autosomal dominant polycystic kidneys
3. • Tuberous sclerosis
4. • Renal vein thrombosis

Hypospadias: % 1:200–300 boys

Definition: External urethral meatus abnormally placed on ventral side of penis

Types:

1. Glandular (glans penis) most common
2. Shaft
3. Perineal "rare but severe"



Varieties of hypospadias.

Rare association: Inguinal hernia – undescending testes.

(NB: Infants with hypospadias must not be circumcised, to preserve tissue for reconstruction)

Rx: Surgical correction at age 1–2 years

Hydrocele:

- Common, usually asymptomatic and sometimes appear blue. They are usually identified during the routine newborn and infant physical examination.
- Accumulation of fluids in tunica vaginalis.
- Transillumination test +ve
- Rx: Most cases resolved by 12 months
- If persistent >18–24 months need repair.


Transillumination test positive

Undescended Testis:

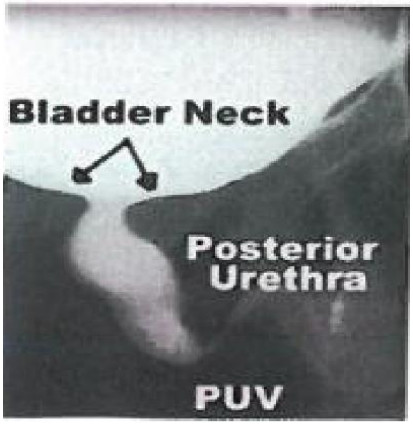
- % 5 in term newborn, more common in PT
- The diagnosis should ideally be made at the routine newborn and infant physical examination of the newborn
- Milking test to differentiate it from rectactile testes.
- If testis undescended but palpable – requires orchidopexy.
- If undescended and impalpable – may require laparoscopy to identify if a testis is present.
- If bilateral impalpable testes – urgent karyotype is essential.
- Complication of undescended testes: increase risk of 1–torsion 2–infertility 3–malignancy.
- Rx: orchidopexy (1year)

Differential diagnosis of empty scrotum: (مهم كلينك Groin examination لجنة)

1. Undescended testes.
2. Ectopic testes (testes palpated below the external inguinal ring but outside the scrotum).
3. Retractile testes (usually do not require surgery)
4. Agenesis.
5. Post orchiectomy.

Posterior Urethral Valves "PUV":

- Is congenital obstruction caused by mucosal fold or membrane obstruct urine flow from the bladder with bilateral hydronephrosis
- Occurs only in male ♂
- Incidence: 1:5000 – 8000 male births
- Most cases are diagnosed on antenatal ultrasound.
- C/P – Poor urinary stream, dripping or UTI
- If not treated early leads to bilateral VUR → Hydronephrosis & pyelonephritis → CKD
- Rx: endoscopic valve ablation – urinary diversion.



Prune –belly syndrome

Prune-belly syndrome (absent musculature syndrome) The name arises from the wrinkled appearance of the abdomen. It is associated with a large bladder, dilated ureters, and cryptorchidism.



Urinary Tract Infection

Types:

1. **Upper UTI**: Acute and chronic **pyelonephritis** (**serious** ??: **kidney** damage +renal scar +CKD)
2. **Lower UTI**: Acute and chronic **cystitis** and **urethritis**
3. **Asymptomatic bacteriuria** (no symptoms + positive urine culture).

UTI in childhood is important because:

1–Up to **50%** of patients have a **structural abnormality** of their urinary tract.
2–Pyelonephritis may damage the growing kidney by forming a **scar**, predisposing to **hypertension** and to progressive **chronic kidney disease** if the scarring is **bilateral**.

Incidence:

- Generally, UTI is more common in **girls** (short urethra)
- About 3%–7% of girls and 1%–2% of boys have at least one symptomatic UTI before the age of 6 years, and 12%–30% of them have a recurrence within a year.
- Neonate and infant upto 6 months of age: –,more common in ♂ (Due to posterior urethral valve)

Predisposing (risk) factors:

1. Femal (Short Urethra)
2. Urinary tract abnormality as
   - I –Vesico – ureteric reflux
   - II –Obstructive uropathy –posterior urethral valve –Pelviureteric junction obstruction
3. Urinary stasis (incomplete bladder emptying)
   - I –Infrequent urination or voiding dysfunction.
   - II–Constipation.
   - III–Neuropathic bladder
4. Uncircumcised boys.
5. Instrumentation (catheterization)
6. Immunosuppression
7. Wiping from back to front in ♀ girls
8. Pinworm infestation and sexual activity

Etiology: (Usually intestinal flora which enter the urinary tract from urethra)

1. **E Coli** (80% most common)
2. Proteus :(More common boys) also (Splitting of urea to ammonia –>Urine alkalinisation)
3. Pseudomonas: especially in children with structural abnormality or long–standing catheterization and neurogenic bladder.
4. Klebsiella :(more in newborn)
5. Others: – Staphylococci, Streptococcus faecalis (**Adenovirus** cause **Hemorrhagic** cystitis)

Route of Infection:

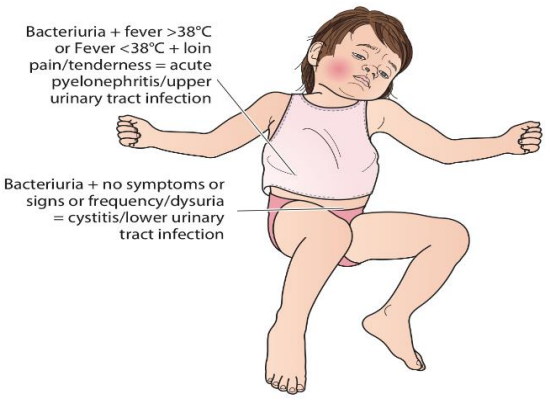
1. Ascending infection (through urethra) (common)
2. Blood–borne: neonatal period.

Clinical Picture: (According to the age)

Neonate & infants (nonspecific)	Older children (more specific)
• Fever • Vomiting, diarrhea. • Lethargy or irritability • Poor feeding/ faltering growth (FTT) • Jaundice (prolonged) • Septicaemia • Offensive urine • Febrile seizure (>6 months)	**pyelonephritis:** • **Fever high grade** with or without **rigors** • Unilateral Loin pain & tenderness. • Nausea, vomiting & diarrhea. • Toxic appearance, dehydration **Cystitis:** • Dysuria, Urgency, Frequency • Suprapubic pain • Hematuria • Incontinence (urine leakage) • 2ry enuresis • Offensive \ cloudy urine

Diagnosis:

1. CBC – Increase WBC (mainly PMN)
2. CRP – High
3. Urine analysis: – Look for urine: WBC, RBC, Nitrite , Leukocytes esterase, bacteria.



Interpretation of results of dipstick testing in children 3 years and older
• Leukocyte esterase and nitrite positive: Regard as UTI • Leukocyte esterase negative and nitrite positive: Start antibiotic treatment if clinical evidence of UTI. Diagnosis depends on urine culture • Leukocyte esterase positive and nitrite negative: Only start antibiotic treatment if clinical evidence of UTI, Diagnosis depends on urine culture • Leukocyte esterase and nitrite negative: UTI unlikely. Repeat or send urine for culture if clinical history suggests UTI

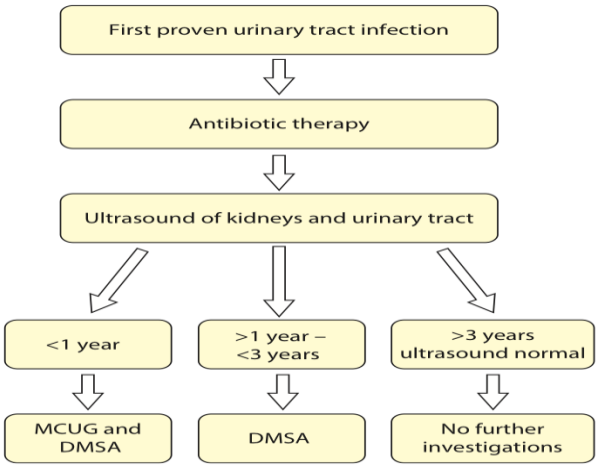
- **urine culture & sensitivity** (growth of single organism per millilitre in a properly collected specimen gives a 90% probability of infection).
    - **SpA**: Any growth CFU is significant (indicate UTI)
    - **Mild stream urine & clean catch:** > (10^5) CFU/ml

4. Blood C & S (in neonate).
5. Urea –creatinine.
6. U/S – Hydronephrosis.
7. MCU: for diagnosis of VUR.
8. DMSA: > 3 months for diagnosis of Renal Scarring.

Extensive investigation done for children who have recurrent UTIs or those associated with atypical features which include:

- seriously ill or septicaemia or poor urine flow.
- abdominal or bladder mass or raised creatinine.
- failure to respond to suitable antibiotics within 48 hours.
- infection with atypical (non–E. coli) organisms.
- Positive family history of renal anomalies.

First urinary tract infection - a protocol for initial management and investigation



Treatment:

- Pyelonephritis
  1. Admission
  2. IVF – Antipyretics
  3. IV antibiotics – (<3m: IV Augmentin or 3rd generation cephalosporin for 7 – 10 days
     - (> 3m & children: IV antibiotic usually for 2–4 days followed by oral antibiotics for a total of 7–10 days)
     - The choice of antibiotic is adjusted according to sensitivity to urine culture.
  4. Follow up
     - BP.
     - Renal growth and function.
- Cystitis
  - Home Rx, Oral Antibiotics.
  - Augmentin, Nitrofurantoin or Trimethoprim for 5 days

Complication:

1–Acute: – Dehydration –abscess

2–Chronic: – Renal Scare – 2ry HTN – CKD

Prevention of UTI:

1. Adequate fluid intake.
2. Frequent & regular voiding (Double micturition).
3. Rx & prevention of constipation.
4. Good perineal hygiene.
5. Antibiotic prophylaxis in VUR grade 1–2–3.
6. Lactobacillus acidophilus.
7. Wiping from front to back in ♀.
8. Circumcision in ♂.

"Proteinuria"

Definition: Is excretion of abnormal amount of protein in urine [normal urine: No and trace amount]

Normal urinary protein excretion:–

- By Urine dipstick: –ve or trace
- In 24 hr Urine protein collection :<150 mg/day
- By U p/cr ratio :< 0.2 %
- Or <4 mg/ m^2/ day

Classification: according to:

- Duration – **Transient** proteinuria or Persistent proteinuria
- Amount – **Heavy** proteinuria or **non–heavy** proteinuria.
- Origin – **Glomerular** as PSGN or Tubular as Cystinosis

Transient Proteinuria:

- Common – self limiting
- Cause: Fever – Exercise – HF –Dehydration – seizures
- Proteinuria just (+ or ++)
- Benign – No renal symptoms
- Rx: reassurance – No evaluation (does not require investigation)

	Heavy Range proteinuria (Nephrotic Range)	Non–Heavy range proteinuria (non–nephrotic range)
Urine Dipstick	+++ or more	+ or ++
24 hr Urine collection	> 3g/day or >40mg/m^2/ day	<3g/day 4–40 mg/ m^2/ day
U P/cr Ratio	> 2%	0.2 – 2%

Orthostatic (postural) Proteinuria:

- Proteinuria occurs only in upright position during the day & it decreases or disappears in the supine position.
- Commonest causes of proteinuria in school age children *
- Etiology – Unknown
- No Renal manifestation – normal Bp – No Edema – normal RFT
- + or ++
- It can be diagnosed by measuring the urine protein–to–creatinine ratio in a series of early morning urine specimens (supine & upright urine collection).

- The prognosis is excellent and further investigations are not necessary needs (follow up – Reassurance)

Nephritic syndrome & Nephrotic syndrome

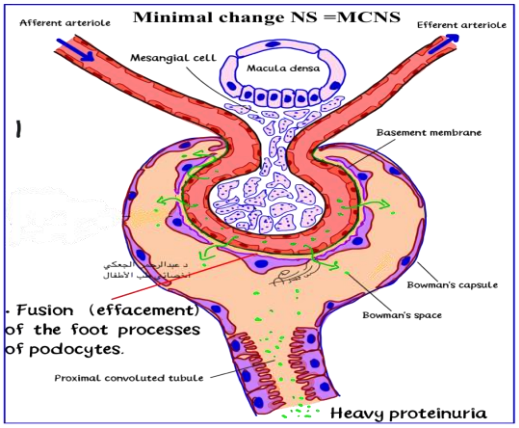
<table>
<tr><th></th><th>NEPHRITIC SYNDROME</th><th>NEPHROTIC SYNDROME</th></tr>
<tr><td>Definition</td><td>is a clinical syndrome of: –
1–Hematuria.
2–Hypertension.
3–Odema .
4–Oliguria.</td><td>is a clinical syndrome of: –
1–Heavy proteinuria.
2–Hypoalbuminemia.
3–Odema.
4–Hyperlipidemia.</td></tr>
<tr><td>Causes</td><td>1–POST STREP GN
2–IgA nephropathy
3–HSP –Lupus nephritis
4–Membranoproliferative GN
5–Alport syndrome

_____-

post streptococcal سيتم شرح موضوع Glomerulonephritis</td><td><table><tr><th>1ry. [95%]</th><th>2ry [≃5%]</th></tr><tr><td>1–Minimal change NS =MCNS (85%)
***
2–Focal segmental GN
3–Membranoproliferative GN
4–membranous GN.</td><td>1–Diseases as HSP – SLE

2–Infections as HBV, HCV, malaria

3–Drugs as
4–Malignancy
5–Allergen as bee sting</td></tr></table>Congenital NS:
%: rare (Finland*)
Onset :1st 3 months
Cause: AR, genetic defect due to nephrin gene mutation (NPHS1)
C/p: large placenta, G oedema –FTT
TTT: nephrectomy & dialysis –late renal transplantation
Prognosis: poor —> CKD</td></tr>
<tr><td>Age</td><td>PSGN (3–12 years) preschool & school age group</td><td>MCNS (2–6 years)</td></tr>
<tr><td>Sex</td><td>more in boys</td><td>more in boys</td></tr>
<tr><td>Patho–physiology</td><td>Group A β hemolytic streptococci (Nephritogenic strain): (throat (serotype 4,12) or skin (serotype 49) —–> latent period —> formation of Ag+Ab +C3 complex —–>deposition in GBM —>Acute glomerular injury:–</td><td>MCNS. (Idiopathic NS)</td></tr>
</table>

	1–Damage to endothelium –> escape of RBC (**COLA URINE =Hematuria** +/–proteinuria + or ++ 2–proliferation of endothelium—>GC obstruction —>↓ Renal blood flow RBF —> + (JGA) –> Angiotensin Renin system > VC –>**HYPERTENSION**. 3–also ↓ Renal blood flow —> ↓ GFR—>fluid retention & hypervolemia —>(**ODEMA**) & (**OLIGURIA**)	H/O URTI —>unknown cause  (altered T lymphocytes function)—>altered GBM –> loss of foot processes —>dysfunction of GFB —> escape of LMW proteins as Albumin –>**Massive Proteinuria–>Hypoproteinemia** (**mainly albumin**)–> • Lead to ↓ osmotic pressure–>Fluid shift from vessels to interstitium–>↓ intravascular blood volume–> + ADH & RAS (aldosterone —> **Generalised edema** • Stimulate liver–> ↑ protein synthesis &↓ lipid catabolism —> **Hyperlipidemia**
Clinical picture	H/O URTI (pharyngitis) 1–3wks or skin infection 3–6wks (impetigo) :– 1–Dark urine (Gross painless cola or tea color) 2–S/S Of HTN [headache – vomiting–dizziness–seizure–LOC – Encephalopathy] 3–edema: orbital –puffy face – anasarca 4–decrease urine output	H/O viral URTI 1–Edema *** Pitting –more in the morning: mainly periorbital–puffy eyes & face /Lower limbs –scrotal or vulval swelling–anasarca (Increased wt) –ascites –pleural effusion. 2–S/S RD (breathlessness)?? Why 3–infection such as peritonitis, septic arthritis, or sepsis due to loss of protective immunoglobulins in the urine. 4–other symptoms (anorexia –malaise – abdominal pain–dizziness–fainting attacks).

<table>
<tr><td></td><td></td><td>

Features suggesting (MCNS) steroid–sensitive nephrotic syndrome are: مهمة

- Age: 2–6 (1–10yrs)
- No macroscopic haematuria
- Normal BP
- Normal blood pressure
- Normal complement levels
- Normal renal function
- Selective Proteinuria
- Renal biopsy: Not indicated
- Good Response to steroids

</td></tr>
<tr><td>Investigation</td><td>

1–**Evidence of GN :–**
Urine analysis:–
Colour – Smoky or gross hematuria
Volume: Oliguria
Specific gravity: High
Proteinuria: – /+ or ++
RBC high –Dysmorphic RBC and RBC & granular casts
2–**Evidence of strep infection:**
THROAT: – ASO Titer high
throat C&S .
SKIN :– Anti DNA as AB +/
Anti hyaluronidase AB + .
3–**C3 level (LOW)**

DD OF LOW C3,C4. Mcq

1. PSGN (low C3)
2. Lupus (SLE) nephritis.
3. Shunt nephritis.
4. MPGN .
5. Subacute IE .

Other investigations:

- RFT(urea –Cr)
- Electrolytes (↓ Na (dilutional), high K (↓ GFR)
- CBC

</td><td>

1–urine sample for **proteinuria**: heavy proteinuria

Urine Dipstick :	+++ or more
24 hr Urine collection :	> 3g/day or
Surface area :	>40mg/m^2/ day
U P/cr Ratio :	> 2%

also look for other urine parameters
2– serum **albumin (low)** < 2,5g/dl
3– lipid profile :**cholesterol (high)** >250 mg/dl.

4–other investigations

- RFT (urea,cr)–Electrolytes –CBC –
- Complement level
- ASO Titer

NB: Low urinary sodium (<10 mmol/L)& high Hb concentration Ht are :– indications of hypovolaemia. [requires urgent ttt with intravenous fluid (0.9% saline or 4.5% albumin solution)]

</td></tr>
<tr><td>Complication</td><td>

PSGN (complex recovery is ≈95% of cases is expected)
1–HTN –>encephalopathy
2–Heart failure.
3–pulmonary edema.
4–hyperkalemia.
5–nephritic nephrosis(rare)
6–ARF /CKD (rare)

</td><td>

1–**INFECTION** (2ry immune deficiency)
Why?

- urinary loss of IG & complements
- S/E of therapy
- Hyposplenism

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<tr>
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<td>
Impaired T cell function
Organisms (encapsulated bacteria as strep pneumonia–E coli –Hib & may be viral as VZV)
Infections: pneumonia–cellulitis–UTI – sepsis –spontaneous peritonitis–chickenpox. (Pneumococcal and seasonal influenza vaccination is recommended, Penicillin prophylaxis is used whilst the child is in relapse)
2–THROMBOSIS: sites anywhere as renal vein thrombosis/pulmonary embolism.
Why?

Loss of antithrombin 3
Increase blood viscosity
Thrombocytosis (+ steroid)
Increased synthesis of clotting factors,Increase fibrinogen concentration.

3–HYPOVOLEMIA: s/s abdominal pain, dizziness, low BP, poor urinary output , fainting attacks.
4–RELAPSE
5–S/E OF MEDICATION (Steroids or other chemotherapy)
6–HYPERLIPIDEMIA (IHD).
7–Renal failure (rare in MCNS)
8–Growth retardation & poor school performances.
</td>
</tr>
<tr>
<td>Treatment</td>
<td>
PSGN >95 of cases —>spontaneous resolution

1–Admission if severe HTN or uncertain diagnosis.

2–Control blood pressure
Diet [low salt diet + restrict IVF]
Drugs [oral or IV according to severity—>diuretics or VD]

3–Eradication of strep infection [penicillin for 10 days]
</td>
<td>
1–Admission especially if

Severe generalized edema
There's complication

2–Nonspecific treatment:
Treatment of edema

[if severe give H albumin 20% iv infusion 1st —then —> give furosemide diuretic IV]
Fluid & salt restriction.
Observation of [BP, WT, input output. clinically improvement]

3–Specific treatment:
STEROIDS [prednisolone oral 60 mg/m2 /day or 2 mg /kg /day]
<table>
<tr><th>Daily dose</th><th>Alternative doses</th><th>Tapering</th></tr>
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</tr>
</table>

كل يوم لمدة 4 أسابيع <- لكي يدخل الطفل في remission	يوم بعد يوم لمدة 4 أسابيع <- للمحافظة على remission	تنزل الجرعة تدريجياً gradually to avoid adrenal crisis . & relapse

other drugs in nephrotic syndrome (alternative therapy)

indication :–

- Steroids resistant NS
- Frequent relapse NS
- Severe steroids S/E
- Relapse on high prednisone dose

Drugs:–

- Cyclosporine A
- Cyclophosphamide
- Levamisole
- Tacrolimus (calcineurin inhibitors)
- mycophenolate mofetil
- Rituximab

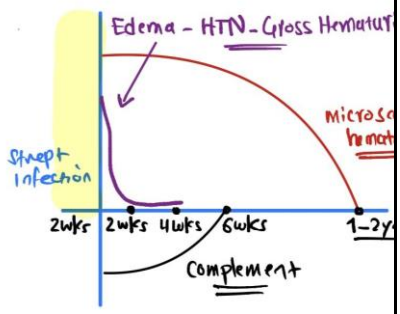
KDIGO مهمة

Remission: – No or trace proteinuria for 3 consecutive days .
Relapse:– +++ or more proteinuria for 3 consecutive days.
SSNS :– remission within 4 weeks of steroid therapy.
SRNS :– No remission > 4–6 weeks of full steroid therapy (60 mg /m2/days)
SDNS :– relapse within 14 days of steroid therapy cessation.
FRNS :– 2 or more relapses /6 months or 4 or more relapses /12 months.

prognosis

Excellent prognosis (complete recovery in ≈95%of cases .

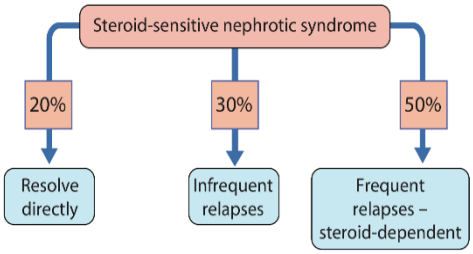
Natural history of PSGN



1. HTN and proteinuria improve within 4–6 weeks of onset
2. C3 normalize by 6–8 weeks of onset
3. Microscopic hematuria may be persistent for 1–2 years

Indication of biopsy in PSGN
1–Absence of low C3
2–absence evidence of strep infection
3–persistent of renal failure
4–persistent microscopic hematuria
5–extra renal manifestation
6–persistent low c3 >3 months

Steroid–responsive nephrotic syndrome



Bad prognostic signs of NS (atypical NS)

1–Age <1 or >10 years
2–persistent HTN
3–Abnormal RFT
4–Gross hematuria
5–Low complement
6–systemic or extrarenal manifestations.
7–Nonselective proteinuria
8– FRNS
9–SDNS –SRNS.
10–Congenital NS.

All these bad prognostic features & signs are also indications of kidney biopsy in NS.

The intensity & duration of of initial CS influence the rate of relapse of NS
—
Idiopathic NS may associate with (Atopy –DM – malignancy as lymphoma)

1. **Minimal change Nephropathy** (♂: ♀ 2:1):
   - By Light microscopy and immunofluorescence: Normal
   - By Electron Microscopy: Fusion (effacement) of the foot processes of podocytes.
   - Usually selective proteinuria (>85% albuminuria)
   - Response to steroid in > 90% of cases
2. **Focal Segmental Glomerulosclerosis** (FSGS):
   - By Light microscopy and Immunofluorescence: Segmental sclerosis
   - By Immunofluorescence: deposits of IgM and C3
   - Resistance to steroids, 30% progress to CKD, recurrent post transplantation.
3. **Membranoproliferative glomerulonephritis:**
   - Fight and electrons microscopy – Increased mesangial cells and < 50% response to steroid.
   - More common in older children., Low complement level.

Steroid–resistant nephrotic syndrome

Table 19.3 Steroid-resistant nephrotic syndrome

Cause	Specific features	Prognosis
Focal segmental glomerulosclerosis	Most common Familial or idiopathic	30% progress to stage 5 chronic kidney disease; 20% respond to tacrolimus, or rituximab Recurrence post-transplant is common
Mesangiocapillary glomerulonephritis (membranoproliferative glomerulonephritis)	More common in older children Haematuria and low complement level present	Decline in renal function over many years Treated with ACE inhibitors and/or immunosuppression with mycophenolate mofetil
Membranous nephropathy	Associated with hepatitis B May precede SLE (systemic lupus erythematosus)	Most remit spontaneously within 5 years

Acute Abdomen + Nephrotic syndrome DxDx

1. Urinary tract infection.
2. Spontaneous bacterial peritonitis
3. Thrombus formation (GIT ischemia).
4. Gastritis and PU (S/E of steroid)

SIDE EFFECTS: mcq

- **Steroids therapy:** → (Hyperglycemia, 2ry DM–Overweight & Obesity (Moon face & buffalo hump)– Myopathy & muscle weakness–Osteoporosis–Skin (poor wound healing, striae)–Infection–Growth Retardation–Hypertension–GIT: Gastric irritation & peptic ulcers–Cataract–Emotional lability, psychosis)
- **Cyclophosphamide:** leukopenia (CBC weekly IF WBC < 5000 – stop drug.)
  Others:(Alopecia –Bone Marrow suppression –Hemorrhagic cystitis– infertility)
- **Cyclosporine A:** S/E (Hirsutism –Nephrotoxicity–HTN– Gum hyperplasia)

Renal Failure

1. Acute Kidney Injury = ARF.

Is

Sudden → Reversible

2– CKD = chronic kidney disease

Is

Gradual → Irreversible

Impairment Of renal function

Acute Kidney Injury:

Types

1. Pre renal AKI (60% most common)	2. Renal AKI (30% most common in UK)	3.Post Renal AKI (10%)
• **Dehydration**** – Most common developing • Hypovolemia (Burns, Hge, Sepsis, nephrotic syndrome. • Heart failure. • Circulatory failure.	• **Vascular** – HUS**, RVT, vasculitis. • **Glomerular** – GN • **Tubular**: acute tubular necrosis, toxin, ischemic • **Interstitial**: interstitial nephritis, pyelonephritis	• Obstructive: as Congenital (PVU) or Acquired stones or blocked urinary catheter.

Clinical Picture:

- H/O Preceding – Cause
  - GE diarrhea – vomiting
  - URTI (APGN)
  - Rash – Arthritis (SLE – HSP).
- S/S of hypovolemia – Prerenal AKI – S/S
  - Dehydration
  - Shock
  - Decrease Bp

If – Renal AKI – S/S (Overload – Edema – HTN)

  - Oliguria – Anuria < 1 cc/kg/hr
  - Acidotic Breathing

Investigation

1. RFT – Ur (high) – Cr (high)
2. Urine Analysis Volume – RBC Cast – RBC
3. ABG: metabolic acidosis
4. Electrolyte: K (high)– Na (low: dilutional)–Ca (low)–ph (high).
5. Ix for the Cause
   - ASO – C3C4 – U/S – ANA
   - Biopsy (RpGN)

	Pre Renal ARF	Renal ARF
Urine Osmolality	High > 500	Low < 300
Urine Na	Low < 20	High > 40
FE Na	<1%	>3%
Specific Gravity	High >1,020	< 1,010

Treatment:

1. **Conservative:** check hydration state
   - Prerenal AKI (Dehydration) – "IV" Fluids (Rx of shock 20ml/kg)
   - Renal AKI (Overload) – Fluid Challenge (diuretics → No Urine →Continue →No Urine →Dopamine (renal dose) + Fluid Balance.

2–Treatment of hyperkalemia (EMERGENCY)

Treatment of hyperkalemia* مهمة :–** under ECG monitoring.

- Ca gluconate 10% IV slowly (Stabilizing the myocardial cell membrane to prevent cardiac arrhythmia)
- Regular Insulin + Dex 50% (to avoid hypoglycemia)
- Salbutamol nebulizer / IV
- Na HCo3 to correct acidosis
- Sodium polystyrene sulfonate orally (Na kayexalate) 1 g/Kg– Oral and enema (chelation)
- Dialysis.

ECG changes in hyperKalemia

1. Prolonged PR interval
2. Wide QRS
3. Tall, peaked T wave*
4. VF
5. Cardiac arrest

3–Treatment of metabolic acidosis by NaHCO3 IV infusion slowly.
4– Treatment of HTN (salt & water restriction –diuretics –other anti hypertensive drugs
5–Treatment of convulsions (ttt the cause: HTN, hypo Na & hypo ca): diazepam, dialysis.
6–**MONITORING:** –Vital signs (BP), Input & Output chart, Hydration state, wt , ABG, RFT, ECG, Electrolytes
7–**Dialysis:** –

Indication of Dialysis in ARF:
1. Hypervolemia: pulmonary Edema and severe HTN due volume overload. 2. Severe hyper K does not respond to conservative Rx. 3. Severe metabolic acidosis does not respond to Na HCo3. 4. Uremic encephalopathy, convulsions & coma. 5. Cr > 10 mg/dl. 6. Severe hypo or hyper Na and Multisystem Failure.

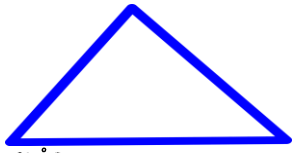
Modalities of dialysis: –

1. Peritoneal dialysis PD: –	2. Hemodialysis HD:–
"Most common modality in infants & small children". • Peritoneal catheter is inserted per–cutaneously or surgically • Hyperosmolar fluid "dialysate" is infused into the peritoneal cavity • Dialysate is allowed to stay for a period 15–60 min "Dwell time", then be drained • The semipermeable peritoneal membrane is used for equilibrium between dialysate & body fluid • This cycle is repeated for 24–48 hours • **Major complications**: peritonitis, hernia, protein loss, respiratory compromise, hyperglycemia	HD is more efficient than PD. • Session is 2–4 hours • Rapid correction of volume overload, hyperkalemia & metabolic acidosis • Vascular access is needed (central venous catheters) **Complications:** **Complications of VA fistula: –** • Insertion: arterial injury, pneumothorax & hemothorax • Infection (local & systemic) • Thrombosis **Complications of the procedure:** • Hypotension. • infection transmission as HCV, HIV. • Heparin (bleeding).

Haemolytic Uremic Syndrome" HUS"

Triad of :–

Haemolytic Anemia "MAHA"
(MicroAngiopathic – Haemolytic Anemia)



Thrombocytopenia ARF

Types:

	Typical HUS	Atypical HUS
Also Called	• Classical HUS • Epidemic HUS	• Non–Classic HUS • Sporadic
H/O Diarrhea	• D + HUS • H/O Bloody diarrhea	• D – HUS • No H/O Bloody diarrhea
%	• 90 – 95% "Common"	• 5–10% "Rare"
Cause	• Infection: vero– cytotoxin–producing E Coli H7 O157 • Shigella, step pneumonia. • Drugs – Disease	• Genetics – Familial (AR)
Pathogenesis	• Endothelial Cells Damage	• Complement Mediated
Prognosis	• Good > 80% Recover • 10 –20%: CKD	• poor > 50%: CKD & ESRD • Eculizumab has greatly improved the prognosis .

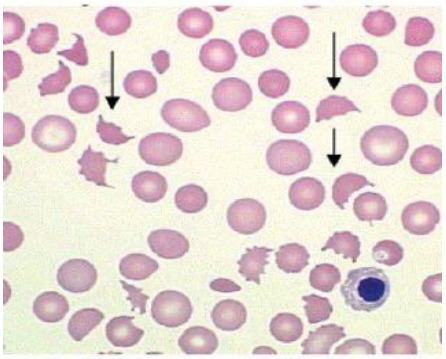
Clinical Picture: (typical HUS) :

The Usual Age 3 – 4 years with H/O Bloody diarrhea for 3–4 days. Then the child:

- Pale – fatigability – irritable.
- Bleeding – Bruises or petechiae.
- Oliguria – Acidosis – seizure, coma

Investigation:

1. CBC: Hb (Low) – Plt (Low) – WBC (N /high).
2. Retic count > 2% (high).
3. Coomb's test (–ve).
4. Urea (high) – Cr (high).
5. Urine Analysis.
6. Blood film (smear): Schistocytes/ helmet cells.



7. Stool analysis –C&S: – E coli
8. PT – PTT (N)
9. U/S – Biopsy (rarely needed)

Treatment:

1. Hospital Admission
2. Treatment of ARI – Fluid – (Dialysis **early**)
3. Treatment of Anemia: PRBCs Transfusion
4. PH transfusion is not indicated
5. Antibiotic is not recommended
6. Thrombolytic is infective

Chronic Kidney disease

Definition: It's irreversible impairment of renal function slowly over months or years

(Either renal injury (proteinuria) and \ or glomerular filtration rate < 60 ml\min\1.73m2 for > 3 months).

GFR (Schwartz formula) = H+ (cm) X K / Ser Cr "mg/dl

Classification of CKD:

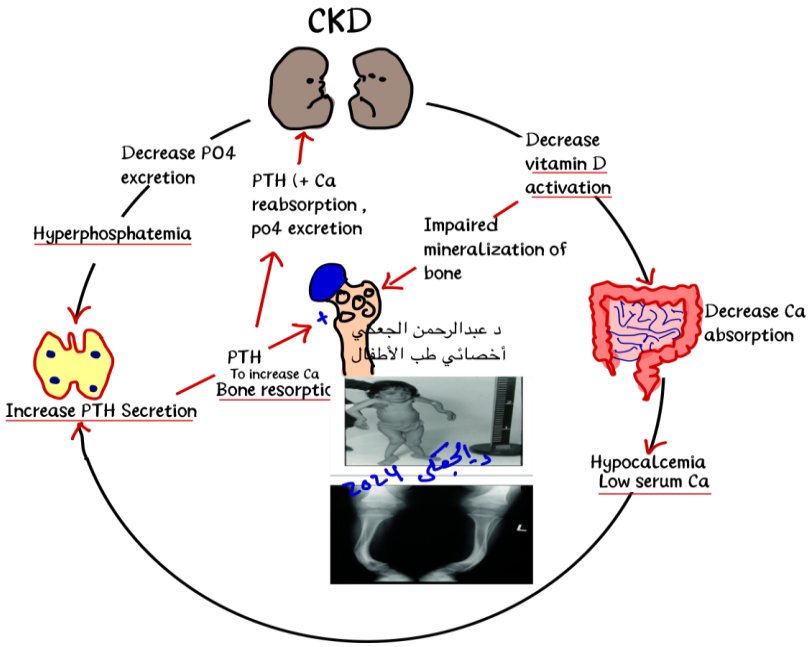
Stages:	GFR (ml/min per 1.73 m2)	Description
stage 1	>90	Normal renal function but structural abnormality or persistent haematuria or proteinuria
stage 2	60–89	Mildly reduced function, asymptomatic
stage 3	30–59	Moderately reduced renal function, renal osteodystrophy
stage 4	15–29	Severely reduced renal function with metabolic derangements and anaemia. Need to make plans for renal replacement therapy.
stage 5	<15	End stage renal failure, renal replacement therapy required

Causes:

1. **Congenital Anomalies** – 40–55 % "most common" (Structural Malformation)	2. **Acquired**	3. **Hereditary or Systemic diseases affecting kidneys**
• Renal hypoplasia • Obstructive Neuropathy PUV – VUR • PCKD • Neurogenic Bladder	• Glomerulonephritis ≃20% • Pyelonephritis • IgA nephropathy • HUS, RVT. • Nephrocalcinosis.	• Alport Syndrome • PCKD • Congenital nephrotic Syndrome • Cystinosis. Systemic diseases such as SLE, DM, SCA, vasculitis

Clinical Picture:

1. **General symptoms** (fatigability –itching –vomiting & abdominal pain [due to gastritis or acidosis] & acidotic breathing, skin pigmentation).
2. Growth retardation (**short stature and FTT**), Why??? Due to: –
   - Anorexia, vomiting (M Acidosis).
   - Resistance to GH.
   - Anemia.
   - Renal osteodystrophy.
   - Electrolyte disturbance.
3. unexplained **normochromic, normocytic anaemia**. Why??? Due to: –
   - Decreased Erythropoietin production
   - Decreased intake of iron and vitamin B12
   - Bone marrow depression by uraemia
   - Defective iron utilization and decrease RBC survivor
4. Bony deformities from renal osteodystrophy (renal rickets). Why ???Due to: –
   - Decrease 1, 25 (OH2) vitamin D3 activation
   - High PO4 ,Low Ca and 2ry hyperparathyroidism
   - Which results in osteitis fibrosa and osteomalacia & fracture mof the bones



5. Hypertension. Why??? Due to: –
   - High Renin level and Na & fluid retention
   - Presented with headache, HTN encephalopathy may leads to cardiomyopathy.
6. Bleeding tendency due Platelet dysfunction
7. Uremic breathing.
8. Delayed Puberty
9. Polyurea and polydipsia (tubular damage: impaired urine concentration) , Nocturnal enuresis
10. Hyperlipidemia, tissue insulin resistance.

Diagnosis:

1. Urea and creatinine: High
2. Electrolyte: –
   - K high: due to (↓GFR, metabolic acidosis & ↑ K intake)
   - Na (Early ↓ Na due to tubular damage late ↑ No due to anuria and ↑ intake)
   - Ca low/ N, PO4 high, ALP ↑, PTH ↑(2ry).
   - Mg high (2ry to decrease renal loss of mg)
3. ABG: Metabolic acidosis Due to
   - ↓ HCO3 reabsorption & ↓ acid excretion by kidney.
4. CBC: Normocytic Normochromic Anemia
5. Renal U/S: Shrunken kidney or any anomalies (Large kidney +CKD in ……..?
6. GFR measurement: assessment of severity
7. Urinalysis: may show proteinuria, hematuria.
8. Echocardiogram, ECG (↑K).
9. Investigation to identify underlying cause as ANA, C3, C4, biopsy.

Treatment of CRF:

I. **Conservative (Symptomatic treatment) (mild to CKD moderate)**

Goal:
1. To Prevent the symptoms and metabolic abnormalities of chronic kidney disease.
2. To allow normal growth and development & school performance.
3. To preserve residual renal function.

- Diet:
  - **High** Calorie intake (Fat and CHO) /recommended daily allowance).
  - **High** Ca diet.
  - **Low** PO4 diet (Restrict milk and milk product) why?
  - **Low** K contains food such as potato, bananas.
  - **Low** salt diet. Why?
  - Ca supplement and vitamin supplementation
  - Protein intake should not be restricted and instead needs to be sufficient (high biological value (meat, eggs, fish.) to maintain growth and a normal albumin.
- Rx of Hyperkalaemia (as discussed in AKI)
- metabolic acidosis by NaHCO3.
- Rx of hypertension
  - Fluid restriction and restrictive salt intake
  - ACE inhibitor , other anti hypertensive drugs.
- Rx of Anemia –
  - Recombinant erythropoietin (S.C) 1–3 times /week

- Iron and folate supplement
- Packed RBC transfusion (in severe anemia)

- Rx of renal Osteodystrophy
  - Dietary phosphate restriction + phosphate binding agents (Ca carbonate, Ca acetate taken with meal)
  - Calcium Orally
  - Vitamin D (Active Form 1alpha) calcitriol (1,25 dihydroxycholecalciferol)
- Rx of growth failure.
  - Correct acidosis and Anemia.
  - Adequate calorie diet.
  - Recombinant GH therapy.

II. **Treatment of end stage renal failure: Dialysis or Renal transplantation– Indication: –**

- Clinical and Laboratory as ARF
- Growth failure
- Pericarditis

"IgA Nephropathy" = 'Berger disease

- The most common chronic glomerular disease worldwide.
- **Age**: any age (Peak 10 – 30 years) increase in Asians and "white" Caucasians
- **Sex**: increase in ♂ (3:1)
- **Pathophysiology**: incompletely Understood (IgA deposition in the glomerulus) Serum IgA Level have no diagnostic value
- **C/P:**
  1. Recurrent episodes of **gross** or microscopic hematuria usually associated with concurrent URTI (1–2 days).
  2. Persistent microscopic hematuria
  3. Proteinuria
  4. HTN
  5. Nephrotic / or nephritic or combined
- **Diagnosis:** By renal biopsy – (mesangial IgA deposition)
- **Prognosis:** most child do not have progressive CKD until adulthood 15 – 20 year after onset of the disease
  Long term follow up is very important.
  The prognosis in children is better than that in adults.
- **Rx**
  1. Blood Pressure control
  2. Dialysis
  3. Renal transplantation

"Alport Syndrome"

- It's most common hereditary nephritis
- Genetic disease caused by mutation in the genes coding for type **IV collagen**
- X linked disease in ≃ 85% (but May AR or AD) , more severe in boys .
- C/P

Kidneys	Ears	Eyes
Nephritis: • **hematuria (earliest &MC)** &/or proteinuria, In Toddler or young children, • HTN, nephrotic syndrome, ESRD 20 – 30 years • Urine analysis, RFT. • Renal biopsy (Mesangial proliferative Glomerular sclerosis. • Genetic test.	• Sensorineural deafness ≃50–75 % bilateral (Progressive with age) • Adiometry	• anterior lenticonus (Pathognomonic)30–40%, corneal erosions & macular flecks • Ophthalmological examination

- Rx
  - Supportive
  - Control HTN– Dialysis
  - Renal transplantation

Nocturnal Enuresis "Bed wetting"

Definition :It is involuntary voiding of urine during sleep after expected age of control .

Usually children are (Dry by day ≃ 2 year) & (Dry by night ≃ 3–5 years)

Epidemiology: More in ♂ 2: 1

- 10–15 % of children at the age of 5 years
- 5% of children at the age of 10 years
- 1% of children at the age of 15 years
- F/H +ve in 50%
- If one Parent was enuretic–>child Risk ≃ 44%
- If two Parents was enuretic–>child Risk ≃ 88%

- Emotional impact to child loss of self esteem, anxiety, poor school performance ..

Types :–

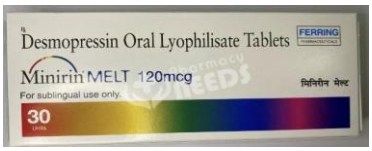
1. **Primary** ≃75% (Common) (when bladder control never achieved)	2. **Secondary** ≃25%(bladder control achieved for 6 M or more & lost)
Causes: 1. Psychological emotional 2. Small bladder capacity 3. Sleeping disorder 4. Hyperactive bladder 5. Genetic factors	**Causes:** 1. Emotional upset * 2. Polyuria: DM – DI, CKD, SCA. 3. UTI, Constipation 4. Seizure ،Obstructive sleep apnea 5. Spinal cord abnormalities

Evaluation:

- History: should be obtained: especially with respect to fluid intake at night and pattern of nocturnal enuresis, Dysuria and frequency. Children with diabetes insipidus, diabetes mellitus, and chronic renal disease may have a high obligatory urinary output and a compensatory polydipsia. also, the family should be asked whether the child snores loudly at night.
- Examination: should include palpation of the abdomen to assess the possibility of a chronically distended bladder. Rectal examination to assess perianal sensation and tone Neurologic and spinal abnormalities.
- Investigation
    a. Urine Analysis (UTI –Glucosuria–Specific gravity)
    b. Blood test – Not routine according to case.
    c. Imaging –U/S and MCU

Treatment:

- <5 year :No Rx
- 5–7 year :motivational counseling +stare chart
- >7 years : Enuresis alarm + Drugs

Motivational counseling	Enuresis alarm	Medication
✓ a–Reassurance of the parents that the condition is self–limited ✓ b–Avoid punishment that may affect the child's psychological development adversely. ✓ c–Fluid intake should be restricted 2hr < bedtime . ✓ d–The child should be encouraged to fully empty their bladder prior to going to bed. ✓ E–A star chart for dry nights. 	✓ –**Enuresis alarm (Conditioning therapy)** ✓ This is a sensor, usually placed in the child's pants or under the child. ✓ Reported success of 60–70%. ✓ Effective, low relapse rate 30% . ✓ Take weeks to the result appear ✓ Often the auditory alarm wakes up other family members and not the enuretic child. ✓ persistence for several months is necessary. ✓ tends to be most effective in older children. 	1–**Desmopressin acetate(DDAVP) 70% success rate .** ✓ ADH analog : that reduces urine production overnight. It is available as a tablet, at bedtime given for 3–6 months . In the past it was used as a nasal spray. ✓ S/E (hypoNa ,seizures)  2–**imipramine (Tofranil)** ✓ a tricyclic antidepressant.. ✓ Reported success rates are 30–60%. ✓ Side effects anxiety, insomnia, and dry mouth ,relapse 60%. 3–**Oxybutynin chloride.** ✓ a pure anticholinergic agent, has been used in some children with primary nocturnal enuresis, but the response rate is LoW, in part because the duration of action is only 6 hr.

❖ Daytime enuresis :is lack of bladder control during the day in a child old enough to be continent (over the age of 3–5 years).
❖ Consider **possible causes**: developmental or behavioural, bladder instability or neuropathy, UTI, constipation, ectopic ureter.

"Henoch – SchonLein purpura"

- Known as :**IgA vasculitis (anaphylactoid purpura)** .
- HSP is the most common Systemic vasculitis in childhood
- HSP it's a small blood vs vasculitis characterized by(purpura– Arthritis – Abdominal pain – Nephritis)
- **Etiology** :Unknown (IgA mediated small VS vasculitis)
- **Age** :3–12 years (peak 4–7yrs)
- **Sex**: more in boys (♂:♀ 2:1)
- **Season**: more in Winter &spring .
- **H/O URTI** present in ≃ 50% of cases .

CLINICAL PICTURE :–

1. Skin	2. Joints	3. GIT	4. Renal
Purpura skin rash (99–100 %) Palpable purpura + petechial rash ,initially start urticarial or maculopapular. **Site**:mainly in the lower limb(extensor surfaces& dependent parts, symmetrical) and bottuck . **Disappear** within weeksThe rash is **first** clinical feature in about 50% and is the **cornerstone** of the diagnosis. 	**Arthritis/Arthralgia : ≃ (75%)** **Large** joints of the lower limb (knee & ankle) Swelling + pain + limitation of movement , refuse to walk. It's **Non** destructing Arthritis symptoms usually resolve before the rash goes. **periarticular oedema** is common . 	**GI tract: ≃ (50–75%)** **Colicky** Abdominal pain **(most common)** Vomiting /Bleeding (gross or occult blood in stool– hematemesis – melena) **Intussusception** is an important complication ≃ 5% 	**Renal: ≃ (40–50%)** **Onset** :either during acute phase or weeks later. Over 80% have **microscopic** or macroscopic haematuria or mild proteinuria. If **proteinuria** is more severe, nephrotic syndrome may result. **Risk factors for progressive CKD** (1–2%) : (heavy proteinuria, oedema, hypertension, and deteriorating renal function) **All children with HSP should be followed** for a year to detect those with persisting haematuria or proteinuria (5%–10%) **BUT** Children who

			havepersistent renal involvement or required treatment for HSP nephritis require long-term follow-up.

Rare Manifestation:

- Scrotal swelling –orchitis– penile edema.
- CNS :ICHge ,headache – Seizures
- Lungs :Interstitial pneumonitis

Diagnosis: "Mainly clinically"

1. CBC :Hb , WBC (N /high) and **Plt is normal or slightly high** in HSP . **{ in case of ITP low plt }**
2. ESR : Normal or high.
3. Urine R/E : Hematuria and Proteinuria
4. Urea /Electrolyte
   - Skin biopsy (shows leukocytoclastic vasculitis, with predominant IgA deposition)
5. Stool for occult blood +ve
6. Coagulation Study :(PT & PTT : Normal)

Treatment:

Mainly **supportive**, HSP (**self-limiting in most of cases**)

1. Bed rest
2. Adequate hydration
3. NSAID for joint pain (Ibuprofen)
4. **Corticosteroid** (prednisone) in sever nephritis and severe abdominal pain, Hge, severe scrotal edema, or CNS involvement, intrapulmonary hemorrhage.
5. **Follow up: Urine analysis & Bp Measurement for 6m after diagnosis.**
6. Immunosuppressive drugs may used in prevention &treatment of kidney disease in IgAV.

Prognosis:

1. Is good in most of cases – recovery within 4 weeks (children have milder disease than adult)
2. Recurrence occurs in ≈ 10 – 20%
3. Follow up of renal involvement in nephrology OPD

"Fanconi Syndrome"

Definition: It's disease of the proximal renal tubules (generalized PT defect) of kidney in which glucose –Na– amino acids – uric acid – phosphate –ca–water – bicarbonate lossed in urine

C/P:

1. Polyuria – Polydipsia – episodes of dehydration
2. FTT – Hypophosphatemic rickets
3. Metabolic acidosis – hypo K – hyper CL –hypo Na.
4. Phosphaturia– glycosuria – aminoaciduria – increase Uric acid in urine

Causes:

1. Congenital: inherited – 2ry to Cystinosis, Lowe syndrome, tyrosinemia, Galactosemia, GSD, Wilson disease.
2. Acquired :Drugs as aminoglycoside or Toxin, amyloidosis .

Rx:

1. IVF (good hydration).
2. NaHCo3 for metabolic acidosis.
3. Give: Alkali,K,Ph,phosphate.
4. Bone: phosphate and active vitamin D

Barrter syndrome	Gitelman Syndrome
• AR • Defect in ascending loop of henle (Na–K–Cl transporter) • More severe • Onset: neonatal(severe) –infancy & early childhood types. • C/P :P+P –FTT –V –Dehydration – constipation –hypotonia. + (Triangular face, protruding ears & large eyes) • Hypo K –metabolic alkalosis • Normal serum mg • Bp N (high R) • High urinary Ca (hypercalciuria & Nephrocalcinosis) • TTT: fluids –K –indomethacin	• AR • Defect in DCT (NaCl cotransporter) • Less severe • Onset: late Childhood & adult • C/P :P+P –FTT –V –Dehydration – constipation –hypotonia • Hypo K –metabolic alkalosis • Low serum mg (high in urine) • Bp N or Low. • Low urinary Ca • TTT: K & mg supplements.

Renal Tubular Acidosis RTA

It is a chronic state of persistent normal anion gap metabolic acidosis (hypokalemic , hyperchloremic metabolic acidosis)due to renal tubular defect .

Types :

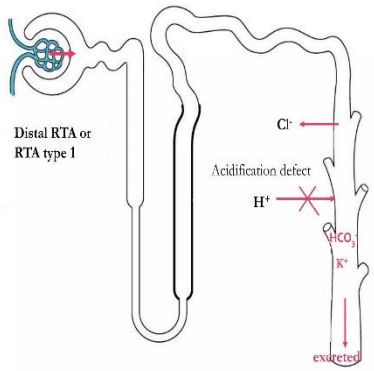
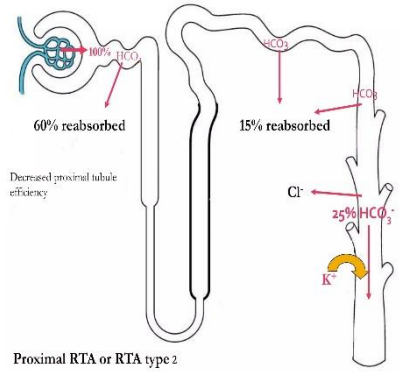
A) Distal RTA: Type I (failure of H excretion by DCT)

B) Proximal RTA=Type II (failure of HCO3 reabsorption by PCT)

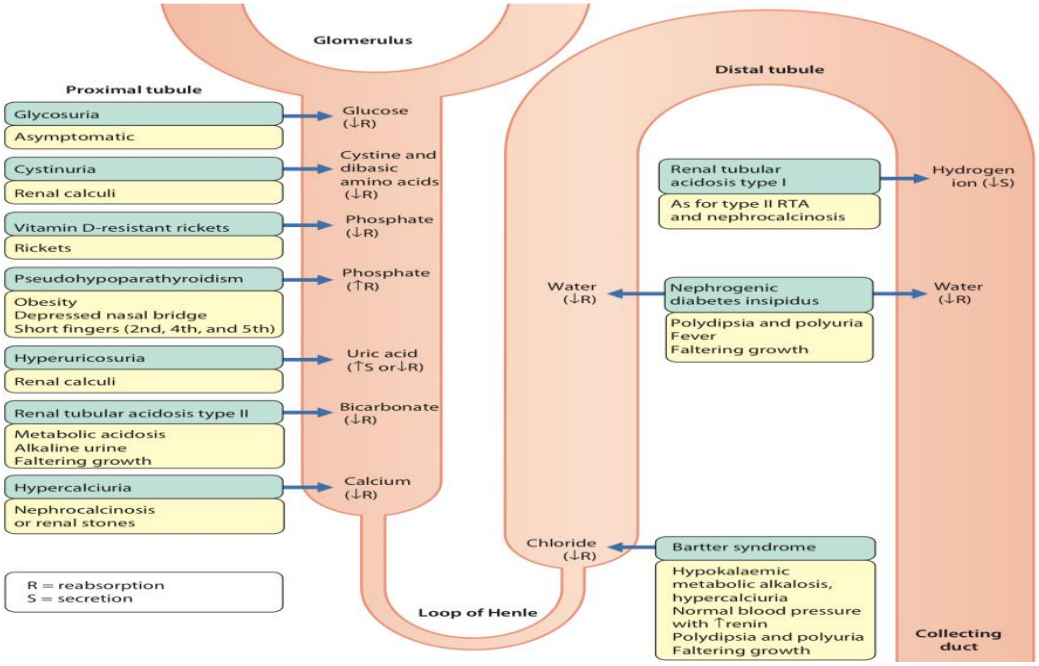
C) Type III (mixed I + II)

D) Type IV (reduced production or unresponsiveness to aldosterone + hyper K)

Comparison of proximal and distal renal tubular acidosis

	Distal RTA (Type 1)	Proximal RTA (Type 2)
Age at presentation	Infancy, childhood (any age)	Infancy
Incidence	Common	less common
Main Pathology	Failure of distal tubular hydrogen ion H excretion 	Failure of proximal tubular bicarbonate reabsorption 
Metabolic picture	Normal Anion gap Hyperchloremic metabolic acidosis with inappropriately high urinary PH	Normal Anion gap Hyperchloremic metabolic acidosis with inappropriately high urinary PH
Urinary PH	Urinary PH never below 5.5 – 6.0	Urine PH can fall below 5.5
Typical presentation (in addition to features of precipitating / causative disease)	• Classically with failure to thrive from infancy Hypokalaemia presents as muscle hypotonia, constipation, hyporeflexia, Polyuria, polydipsia.	• Classically with growth failure and vomiting. Hypokalaemia presents as muscle weakness hyporeflexia, Polyuria, polydipsia

	• Renal stones (nephrocalcinosis). • Rickets secondary to bone buffering the acidosis.  	• in addition, features of Fanconi syndrome may be present
Hypokalaemia	++	+
Treatment	Aim to correct acidosis to allow normal growth and prevent nephrocalcinosis potassium and bicarbonate supplements. Beware of Correcting acidosis before potassium supplementation because of possible life threatening hypokalaemia	Aim to correct acidosis to allow normal growth, Potassium and bicarbonate supplements
Bicarbonate replacement	Small doses such a 2–4 mmol/kg/day	Large doses such as 10–20 mmol/kg/day
Prognosis	Variable	Good



Type IV (reduced production or unresponsiveness to aldosterone)

End organ target failure or low aldosterone:

- Loss of sodium –>hyponatremia
- Retention potassium –> **hyperkalemia**
- Absorption of chloride –> hyperchloremia
- Decreased excretion of acids –> metabolic acidosis
- Loss of fluid –>dehydration

Acidosis & anion gap " mcq "

Anion gap=[Na] – [Cl + Hco3]
< 12 = **normal** or absence of anion gap
Causes: {**diarrhea, RTA, carbonic anhydrase inhibitor, ureterosigmoidostomy**}

>20 =**high** anion gap
Causes: {**lactic acidosis, DKA, inborn errors of metabolism, uremia AKI,CKD, poisoning with (salicylate, methanol, ethanol)**}

Wilms tumor & Neuroblastoma

	Wilms Tumor	Neuroblastoma
Origin	From embryonal renal Tissue "nephron blastoma"	From neural crest cells (adrenal – neck – pelvis)
%	Most common renal Tumor ≃ 6%	Most common extra cranial solid tumor ≃ 8–10% Most common abdominal tumor <1 year
Association	sporadic or Associated 1. Aniridia (no iris) 2. Beckwith wiedman syndrome 3. WAGR – Wilms , Aniridia , GU anomalies, Retardation mental 4. denis–drash syndrome	1. Sporadic 2. Familial 1–2 %
Age	1–5–year peak (>80% of case < 5 years) ♀= ♂ More in Black	Birth – 5 years (infancy) More in ♂ More in white
Clinical picture	1. Abdominal mass (most common >50%) :incidentally in an otherwise well child. and never cross midline 2. Hematuria Painless 3. HTN,anorexia,Abdominal pain. 4. Anemia (haemorrhage into mass) 5. Sign of metastasis to lungs (RD)	1. Abdominal mass 40% , cross midline 2. General symptoms: weight loss– vomiting, poor feeding – pallor fever. 3. Increase CA irritable – diarrhea – Sweating – high Bp, high HR – Flushing 4. Metastasis ≃ 50% (bone BM): Pancytopenia and bone pain ,limb ,paraplegia, . 5. eyes: Raccoon eyes, proptosis 6. skin: nodules. 7. Opsoclonus myoclonus (or dancing eye syndrome) good prognosis
Site	Usually Unilateral ≃ 5–10% Bilateral (Synchronous)	Unilateral

Investigation	1. U/S Abdomen 2. CT – MRI Abdomen: ≃10% calcification 3. CBC – U/Es – CX Ray 4. Nephrectomy & biopsy.	1. U/S Abdomen 2. CT – MRI Abdomen (staging) 3. Urine for (VMA/HVA): High 4. CBC – BMA 5. LFT – U/E – LDH 6. Biopsy is confirmatory
Treatment	1. Chemotherapy 2. Surgery (nephrectomy) 3. Radiotherapy (in advanced cases)	Multidisciplinary 1. If localized – surgery. 2. If old child + mets Chemotherapy, surgery Radiotherapy 3. Relapse is common
Prognosis	Good ≃ 80% cure rate	It depends on 1. Age 2. Stage of disease >1 year + advanced: Poor prognosis 3. Biological character

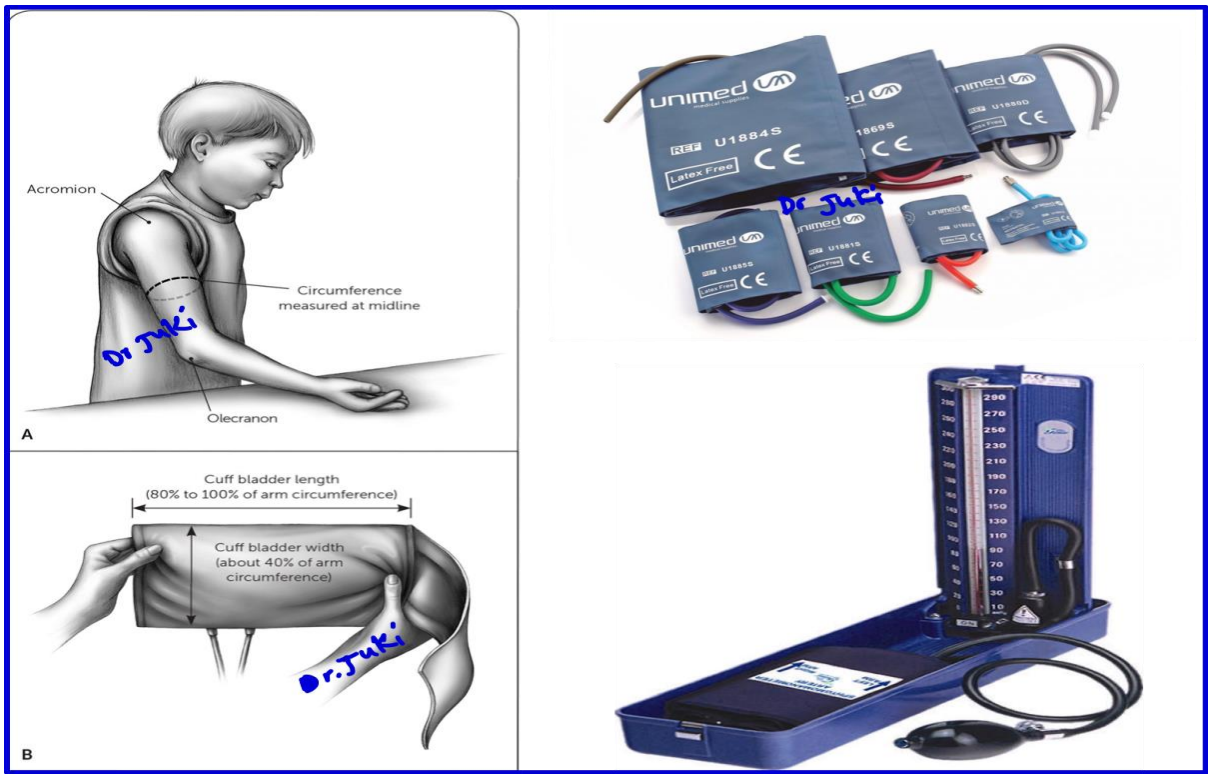
Hypertension in children

Definition:

Category	Systolic or diastolic blood pressure percentile
Normal Prehypertension Stage 1 hypertension Stage 2 hypertension	<90th ≥ 90th and < 95th ≥ 120/80 mmHg independently of the 90th percentile value in adolescents ≥ 95th and < 99th+ 5 mmHg ≥ 99th+ 5 mmHg

- Hypertensive emergency? & hypertensive urgency?
- Factors affecting Bp in children:
  1. Age, gender.
  2. Height, obesity, diet
  3. Genetic factors (familial)
  4. Birth weight (LBW increase Bp)

- BP measurement: for accurate measurement of BP:
  - Proper cuff size.
  - The child should be relaxed and comfortable.
  - The machine, heart and limb should be at the same level.



Blood pressure

Table A.1 Systolic blood pressure for males and females by age and height centile. In children 1–13 years: elevated BP is >90th centile, hypertension (HTN) is ≥95th centile or ≥130/80 (whichever is lower). In young people ≥ 13 years: elevated BP is ≥120/80, HTN is ≥ 130/80.

		Systolic BP (mm Hg)					
		50th height centile		90th height centile		95th height centile	
Age (years)	Height centile	Males	Females	Males	Females	Males	Females
1	50th	86	86	88	88	88	88
	90th	100	100	101	102	101	102
	95th	103	103	105	105	105	105
5	50th	94	93	96	96	96	96
	90th	106	107	108	110	108	110
	95th	109	110	112	113	112	113
10	50th	100	99	102	102	103	103
	90th	112	112	115	115	116	116
	95th	116	116	120	119	121	120
12	50th	104	105	108	108	109	108
	90th	117	118	121	122	122	122
	95th	121	122	126	125	128	126
15	50th	113	108	114	109	114	109
	90th	128	122	130	123	130	124
	95th	132	126	135	127	135	128

Types and Causes:

Primary (essential) HTN	Secondary HTN
Rare in pediatric ≈10% More in adolescents	more common ≈ 90% 1. Renal Disease – **Most common** Renal parenchymal disease GN – Pyelonephritis, CKD or Renal vascular disease as RAS – RVT 2. Endocrine disease as Cushing syndrome , Hyper T3T4 ,Pheochromocytoma 3. Cardiac disease – COA 4. CNS as GBS 5. Drugs as Steroids

Examination:

1. General: Built – Obese – thin – short
2. Dysmorphic Features: Turner – cushing
3. CVS Exa : Pulse – RFD
4. Abdomen: Renal Bruit
5. Genitalia: Ambiguous genitalia

Investigation:

1. U/E, Urine analysis
2. U/S, echocardiogram – Doppler.
3. ABG, TFT
4. Biopsy renal

5. Other Ix (plasma renin activity –aldosterone)

Treatment:

1. General Measures
   - Weight reduction (if obese)
   - Increase physical activity
   - Salt restricted diet
2. Drugs : ACEIx –diuretic –hydralazine– CCB
3. Rx Underlying Causes

Renal calculi (Stones):

Renal stones are **uncommon** in childhood

predisposing (risk factors):

1. • **UTI**
2. • **structural anomalies of the urinary tract**
3. • **metabolic abnormalities.**

- The **most** common are **phosphate** stones associated with infection, especially with **Proteus.**
- **Calcium**–containing stones occur in idiopathic hypercalciuria, the most common metabolic abnormality, and with increased urinary urate and oxalate excretion.
- Deposition of calcium in the parenchyma (nephrocalcinosis) may occur with hypercalciuria, hyperoxaluria, and distal renal tubular acidosis.
- Nephrocalcinosis may be a complication of furosemide therapy in the neonate.
- Cystine and xanthine stones are rare.
- Clinical picture: (haematuria, loin or abdominal pain, UTI, or passage of a stone).
- Investigation: urine analysis–Urine pH, Ca, PO4, oxalate, cystine, uric acid, citrate, creatinine– blood test –RFT –CT Scan*.

Treatment :

- surgical either lithotripsy or surgery,
- A high fluid intake is recommended in all affected children.
- If the cause is a metabolic abnormality, specific therapy may be possible.

DD of Scrotal swelling:

Painful	Painless
1. Testicular Torsion 2. Torsion of appendix testis 3. Epididymitis 4. Trauma 5. Inguinal hernia 6. Idiopathic scrotal Edema (Incarcerated)	1. Hydrocele 2. Inguinal hernia 3. Varicocele 4. Spermatocele 5. Testicular Tumor

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4- www.uptodate.com

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تنبيه:

هذا العمل تم بعد جهد كبير من قبل الدكتور الاخصائي: عبدالرحمن الجعكي، ويعتبر ثمرة سنين من خبرة ممارسة طب الأطفال ، كل ذلك كان ممزوج بحنكة العطاء الاكاديمي للمادة

لهذا لايجوز نسخ أو اقتباس أي جزء من المحتوى الآ بعد التواصل مع الناشر: الدكتور عبد الرحمن الجعكي.